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Problem Solving Protocol

oncoPredict: an R package for predicting *in vivo* or cancer patient drug response and biomarkers from cell line screening data

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Abstract

Cell line drug screening datasets can be utilized for a range of different drug discovery applications from drug biomarker discovery to building translational models of drug response. Previously, we described three separate methodologies to (1) correct for general levels of drug sensitivity to enable drug-specific biomarker discovery, (2) predict clinical drug response in patients and (3) associate these predictions with clinical features to perform *in vivo* drug biomarker discovery. Here, we unite and update these methodologies into one R package (oncoPredict) to facilitate the development and adoption of these tools. This new OncoPredict R package can be applied to various *in vitro* and *in vivo* contexts for drug and biomarker discovery.

Key words: drug discovery; drug response prediction; biomarker identification

Introduction

High-throughput cancer cell line screening datasets are important resources for drug discovery [1]. The two largest publicly available screening efforts are the Broad Institute's Cancer Therapeutics Response Portal (CTRP) [2] and Sanger's Genomics of Drug Sensitivity in Cancer (GDSC) [3]. By providing multi-omic cell line features as well as summarized drug response information, these screens enable identifying associations between a compound's activity and genomic features. These data have become important tools for both biomarker discovery and for developing predictive models of drug response (reviewed in [4–9]). We previously published work using cell line screening data for robust biomarker discovery [10], drug response imputations [11] and subsequent discovery of novel biomarkers from drug response imputations [12].

Here, we offer a new R package, oncoPredict, that provides a convenient wrapper for these three separate methodologies (Figure 1). oncoPredict enables easy implementation of each of the three methodologies, with each pipeline being summarized into a single R function: GLDS, calcPhenotype and IDWAS, respectively. calcPhenotype allows for fitting and predicting drug response based on baseline transcriptomic cell line data [4]. This functionality is similar to our original pRRophetic package [13] but has been overhauled to enable faster implementation, provides updated CTRP and GDSC screening data and allows for increased user customization. oncoPredict also integrates two methods of biomarker identification. The first method, imputed drug-wide association study (IDWAS), identifies biomarkers directly in clinical data based on associations of predicted drug sensitivity with copy number variation (CNV) and other somatic mutation data [12]. The second method performs biomarker

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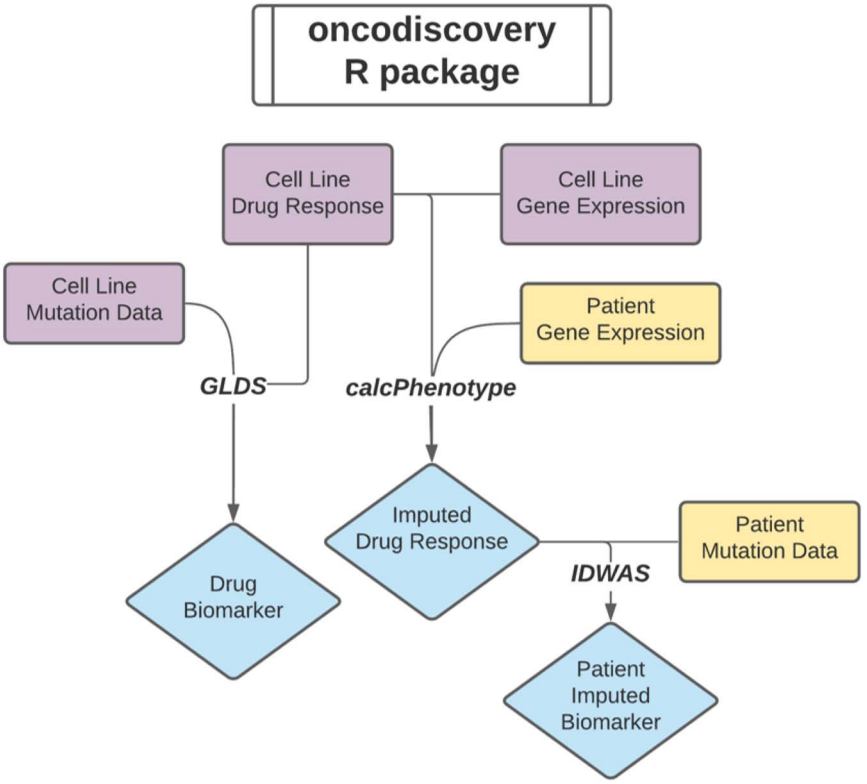

Figure 1. Overview of the oncoPredict R package. Flow diagram showing the inputs (rectangles), the functions (lines) and outputs (diamonds) from oncoPredict. This covers the three primary functionalities included in the package for drug response and biomarker prediction. Purple and yellow indicate cell line and patient input data—these inputs are anticipated to be the most common for the associated function, but the functions are flexible and capable of accepting data from other types of biological systems as discussed in the text.

discovery directly in the cell line dataset and aims at identifying drug-specific biomarkers after accounting for variability in general levels of drug sensitivity (GLDS) [10].

By integrating these methodologies together into one R package, we enable easy implementation of these computational drug discovery tools. While previously described, implementing these methods became inaccessible to many potential users due to difficulty of finding the code, inflexibility of the code for adapting to new uses as well as R version updates. This R package, however, provides updated and convenient implementation of each pipeline. Additionally, integrating these three methods will allow users to perform simultaneous biomarker and imputations for drug discovery, similar to that done in [14], which could increase the impact of their computational discoveries.

oncoPredict bridges *in vitro* and *in vivo* data by expanding the utility of existing *in vitro* data to enable *in vivo* discovery. The primary use cases for each function are summarized below. Vignettes are provided in our Github repository to cover special use cases. All R source codes are publicly available via GitHub and on our website (see Availability section).

Methods and results

To facilitate usability and robustness of oncoPredict, the transcriptomic and cell line response data from CTRP and GDSC have been downloaded, processed and included. For CTRP, drug response data were obtained from the Cancer Target Discovery and Development Network established by the National Cancer Institute's Office of Cancer Genomics [15]. The corresponding gene expression data are hosted by the Broad Institute's Cancer Cell Line Encyclopedia (CCLE) data portal [16]. The GDSC gene expression and drug response values were downloaded from the GDSC website [17]. GDSC consists of two datasets, GDSC1 and GDSC2 dataset, and the data from both are included in oncoPredict. These data are together referred to as GDSC in the remainder of this paper. After download, the data were processed and formatted into gene expression or drug response data matrices.

GLDS

Method principle

We previously reported on a GLDS phenomenon and its effect on biomarker identification. GLDS is the observation that individuals in a population (cell lines or patients) can be, in general, more sensitive, or more resistant to a therapy, no matter the treatment. As shown in the original paper, this phenomenon is related to, but not necessarily the same as, multi-drug resistance (MDR). Correcting for this variable allows for biomarker identification that is more specific to the drug of interest.

To correct for GLDS, for a given drug, a set of negative control (i.e. completely unrelated) drugs are selected. These negative control drugs are unrelated mechanistically and have a measured cellular response that is not highly correlated ($r_s < 0.7$) with the drug of interest. GLDS is then estimated as the first 10 principal components (PCs) of the set of negative control response values in a panel of cancer cell lines. These PCs were then included as covariates in a regression model for drug sensitivity

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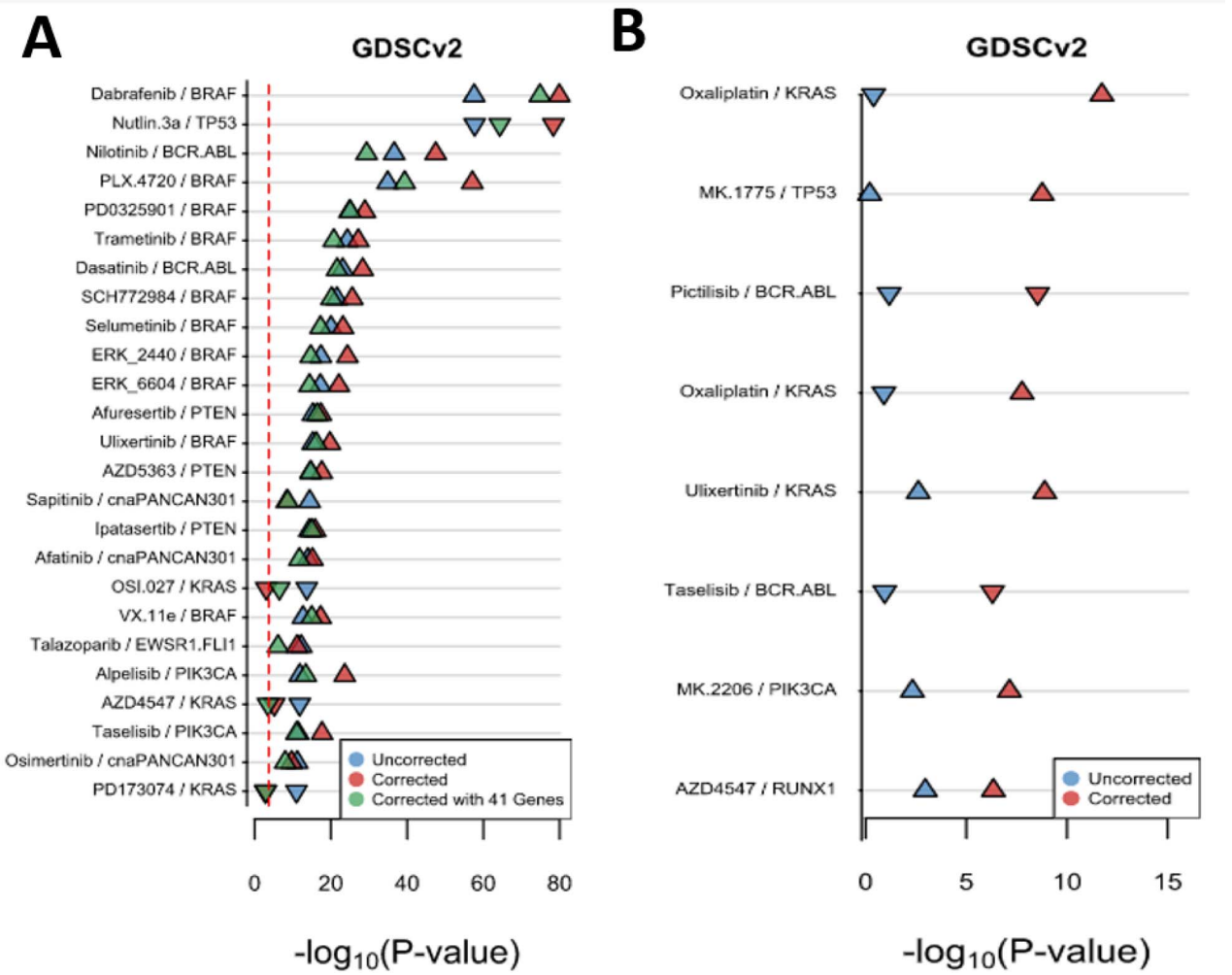

Figure 2. Correcting for GLDS Improves Biomarker Identification in GDSCv2. (**A**) Dot plot of top gene–drug associations with and without correction for GLDS. Also included is the significance of the associations after correcting for GLDS using a gene set derived from the data. The direction of the triangle indicates the direction of the association: triangle pointing up indicates the drug is more effected in the mutated setting. Red dash line represents significance. (**B**) Similar graph as in (**A**), but graphed are gene–drug associations that were not significant in the original data (blue) but were significant after correcting for GLDS.

against gene mutations with the goal to identify drug-specific biomarker(s). If negative control drugs are not available for a compound, a gene signature can also be utilized to correct for GLDS. This gene signature is a way of estimating GLDS without dependence on large-scale drug screening datasets.

Function utility and updates

The GLDS function identifies GLDS uncorrected and corrected associations between drug sensitivity and a phenotype of interest, like somatic mutation. We have included the outputs obtained from applying the function to several large-scale drug screening datasets (including GDSC and CTRP) in oncoPredict's Open Science Framework (OSF). Also included in oncoPredict's OSF are the data required to generate these associations, including updated GDSC and CTRP drug relatedness and CNV and coding variant data for pan-cancer. This output includes all drug–gene relationships which have been uncorrected and corrected for GLDS. The GLDS function has been modified to improve usability when applied to datasets beyond GDSC and CTRP. We also modified the function to give the user the ability to adjust the relatedness parameter, selecting the threshold for high correlation when filtering negative control drugs. This provides a more reliable method of filtering negative controls than what was previously employed in GLDS.

In addition, one can now generate a gene signature as a surrogate of GLDS using the package by providing GLDS function with a gene expression profile. The expression of these genes is significantly correlated to each of the 10 PCs used to estimate GLDS across a drug screening dataset. In other words, they also represent negative controls. The performance of both the PC method and the gene signature method is shown in Figure 2 using GDSCv2 data.

Example use case

We originally showcased this method by correcting for GLDS in Sanger's GDSC 2014 screening data. Here, we applied our package to identify biomarkers of response with and without controlling for GLDS in the separate GDSCv2 dataset (Figure 2). In addition, we used the GLDS signature functionality to generate a 41-gene signature from this dataset, consisting of genes highly correlated with GLDS. Thus, the package enables two ways of conditioning on GLDS; one is through the top 10 PCs, which is implemented

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in the GLDS function, and the other is with a 41-gene signature, which users can use as a way to correct for GLDS without the use of additional large-scale drug screening data. Consistent with previous results, genes associated with MDR, such as CYR61, were included in this signature.

Figure 2A shows the top gene–drug associations with and without correcting for GLDS. The gene–drug associations shown here were corrected for GLDS both by directly estimating GLDS from the data and by estimating using the 41-gene signature set to show the concordance of the two methods and to provide an estimation for GLDS. In other case, significant gene–drug association disappears after being conditioned on GLDS. For example, PD173074 (a FGFR inhibitor) was predicted to have a significant association with *KRAS* mutations without GLDS correction; however, this association was insignificant with correction for GLDS. These observations support previous findings that controlling for the first 10 PCs or gene signature can be used to describe GLDS. In our previous work, we showed that correcting for GLDS reduced spurious findings. Similar results are shown here. On the other hand, some meaningful associations can only be found when conditioned with GLDS as shown in Figure 2B. For example, KRAS is an established biomarker for oxaliplatin [18], but this association was only identified after correcting for GLDS.

Predicting drug response function: calcPhenotype

Method principle

This function implements the pipeline for the prediction of clinical chemotherapeutic response by using only baseline tumor gene expression data. A complete technical description of this pipeline is described in [11]. Briefly, large-scale gene expression and drug screening data (training dataset) are used to build ridge regression models that can then be applied to new gene expression datasets (testing dataset) to yield drug sensitivity predictions for the new dataset [19]. These drug models are built following removal or summarization of gene duplication, homogenization (batch correction) and filtering of low-varying genes. In oncoPredict, users can use preset standards that have been identified depending on the type of data presented (microarray, RNA-Sequencing, etc.; see vignettes) or specify summarization, homogenization and filtering parameters. Finally, our calcPhenotype function is applied to the processed, standardized and filtered clinical tumor expression data, yielding a drug sensitivity prediction for each patient.

Function utility and updates

This calcPhenotype function can be used in retrospective studies to predict drug response and compare with the observed patient response to investigate model performance. Recently, however, we showed that patient-imputed drug response can be used for drug discovery directly in the patient data [14]. That is, by looking for differences in the imputed drug response values between two related cancer subsets, we could identify compounds showing higher efficacy toward one cancer subtype. Finally, this functionality can also be used to quickly turn any patient transcriptome dataset into a dataset for biomarker discovery as discussed with the IDWAS function.

Users can either use the GDSC or CTRP data that are prepackaged into the oncoPredict as the training datasets or can supply their own training data in order to predict drug response in any gene expression matrix provided. Genes that are duplicated in either the training or testing datasets are removed or summarized by the mean expression based on user input. Vignettes describe the optimal options for preprocessing the training and testing expression data for microarray, RNA-Seq or a mix of both. Compared to the method published with the original manuscript, the calcPhenotype function was updated to give users more options for model generation and evaluation. The default predictive model provided is ridge regression, but principal component regression (PCR) is also available in calcPhenotype as this method also performs well in drug response predictions [20]. Regression evaluation metrics (R^2) can easily be obtained for each model. Lastly, the function was updated to provide an option to calculate the correlations between predicted drug response and gene expression as a way of identifying expression-based biomarkers of drug response.

Example use case

To assess the model performance of the calcPhenotype function, we imputed drug response in an ovarian cancer clinical dataset, which had both patient response to paclitaxel and gene expression microarray data (GSE51373) [21]. We used the CTRP dataset to train a model of paclitaxel drug response. We then stratified patients by their observed clinical outcome (defined by the trial as resistant or sensitive to paclitaxel treatment) and compared the predicted sensitivity scores (Figure 3). We see that imputed paclitaxel sensitivity is able to correctly stratify patients into their responder/non-responder categories ($P = 0.0075$ by *t*-test). Additionally, the area under the receiver operating characteristic (ROC) curve was 0.79, which is consistent with the performance of the models of the original paper [11].

Identifying biomarkers from patient data: IDWAS

Method principle

The IDWAS approach is an extension of the imputing drug response values that readily enables biomarkers identification in that population. IDWAS is similar in conception and implementation to GWAS. That is, associations between imputed drug response values and either somatic mutations or CNV are determined by using linear models in R in order to estimate drug–gene interactions and identify biomarkers of drug response. By taking advantage of the large sample size of clinical datasets, this allows for the identification of novel relationships that are not seen in cell line datasets. Additionally, since it uses patient genomic data, any biomarkers found are pertinent to that patient population. The full methodology for the imputed drug-wide association study (IDWAS) is described in [12].

Function utility and updates

To outline the method employed by the IDWAS function, the function takes two inputs: a matrix of patient genetic features and a matrix of imputed drug response values (e.g. the output of the calcPhenotype function). The genetic features can either be a mutation matrix containing 0/1 to describe wild-type/mutated gene status or a CNV matrix containing the CNV levels of the genes in patients. The IDWAS function allows flexible usage of either type of data. Additionally, by integrating the IDWAS and calcPhenotype functions into a single package, the use of the IDWAS methodology is greatly facilitated. Users are able to easily obtain drug response predictions in

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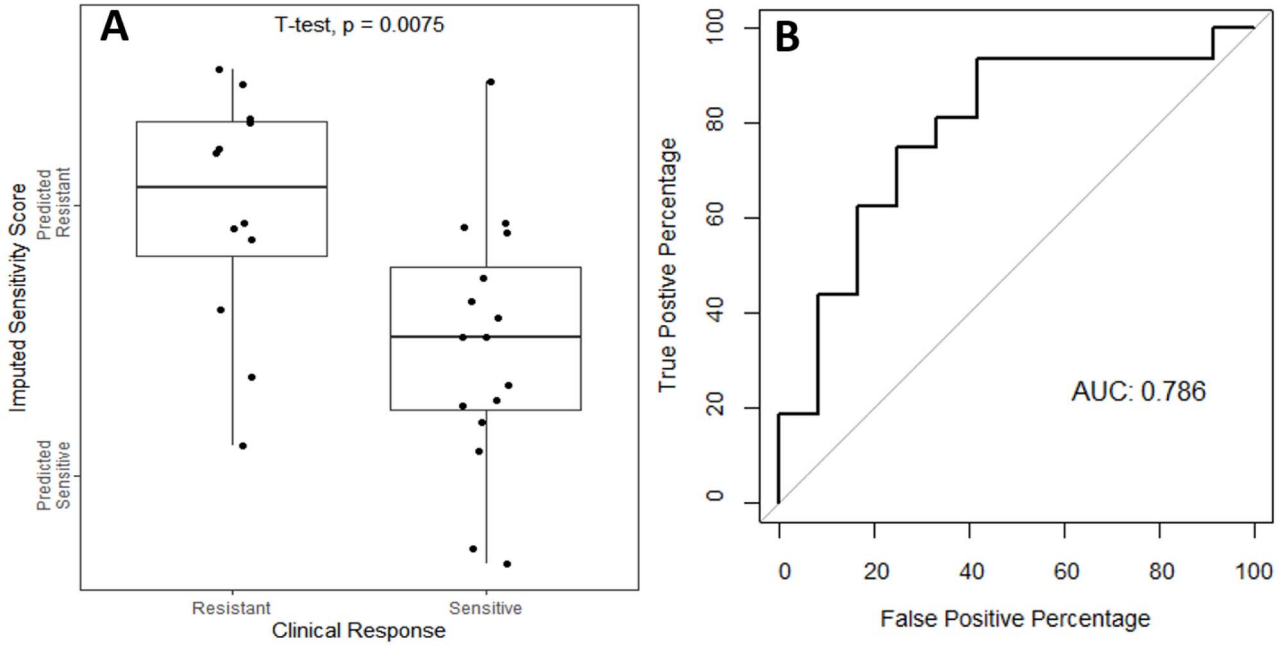

Figure 3. Paclitaxel imputed response in ovarian cancer clinical trial (GSE51373). (**A**) Patient expression data were downloaded from GEO, processed and then used as a testing expression data for the oncoPredict calcPhenotype function. Actual clinical outcomes for the patient were then plotted against the imputed drug response output (*y*-axis, a continuous response value). (**B**) ROC curve showing the percentage of true positives against the percentage of false positives as the classification threshold is varied.

any clinical dataset by using calcphenotype and then search for biomarkers by using IDWAS. Vignettes are provided to demonstrate how to download and apply this methodology to the TCGA datasets, as described in the below use case. To aid in usability, pan-cancer TCGA mutation and CNV matrices are provided in oncoPredict. However, users could easily adapt this application to other datasets like the International Cancer Genome Consortium or cancer-specific datasets such as METABRIC.

Example use case

The Cancer Genome Atlas (TCGA) [22] gene expression data were downloaded from firebrowse.org, and mutation data were downloaded using the TCGAbiolinks R package [23]. TCGA mutation data were summarized on a per-gene basis by calling a gene mutated if the protein had any mutation that would affect the amino acid sequence, which was formatted into a mutation matrix. For the genes assessed in this paper, genes mutated with a frequency above 0.5% across all patients were included.

Using oncoPredict's calcPhentope function, we built drug response models with the CTRP data. We imputed drug response for all 496 drugs across the TCGA samples ($n = 8536$). Similarly, to our previous report [12], the top significant association was that between nutlin-3a (an MDM2 inhibitor) and *TP53* status. As can be seen in Figure 4**A**, this is a highly significant and specific association. The direction of this association (not shown) is consistent with expectations and indicates that *TP53* mutations render nutlin-3a ineffective. Conversely, for selumetinib (a MEK inhibitor), the association was highly and specifically associated with being more effective in the *KRAS* mutated setting (Figure 4**B**).

Road map for future updates

To ease use, we have downloaded and packaged the most up-to-date CTRP and GDSC data, two of the largest publicly available drug screening datasets. The package will be updated twice yearly to include any major updates to these data. Additionally, we will incorporate new and appropriate screening data resources as they become available.

Further feature requests and ongoing changes to increase usability will continue over the life of the package. For example, we have plans to further integrate calcPhenotype, IDWAS and GLDS to make the computational discovery of biomarkers and drug response predictions even more seamless based on user feedback. By integrating these distinct but separate methodologies, the package will give users a one-stop shop instead of the previously dispersed and disaggregated methods linked to individual papers.

The pipelines implemented in the package are 'state-of-the-art' and represent accurate and convenient drug response prediction methods. In general, however, the field lacks guidelines for the best use and application of drug response prediction methods. Similar to the inclusion of PCR as an option in the calcPhenotype function, as we and others in the field identify gold standards for drug response predictions, oncoPredict, will continue to update.

Conclusion

In conclusion, we have presented the R package oncoPredict that bridges the *in vitro* drug screening with *in vivo* drug and biomarker discovery. One can easily predict patient tumor response to a large number of drugs screened in cancer cell lines and can perform biomarker discovery both with and without GLDS condition.

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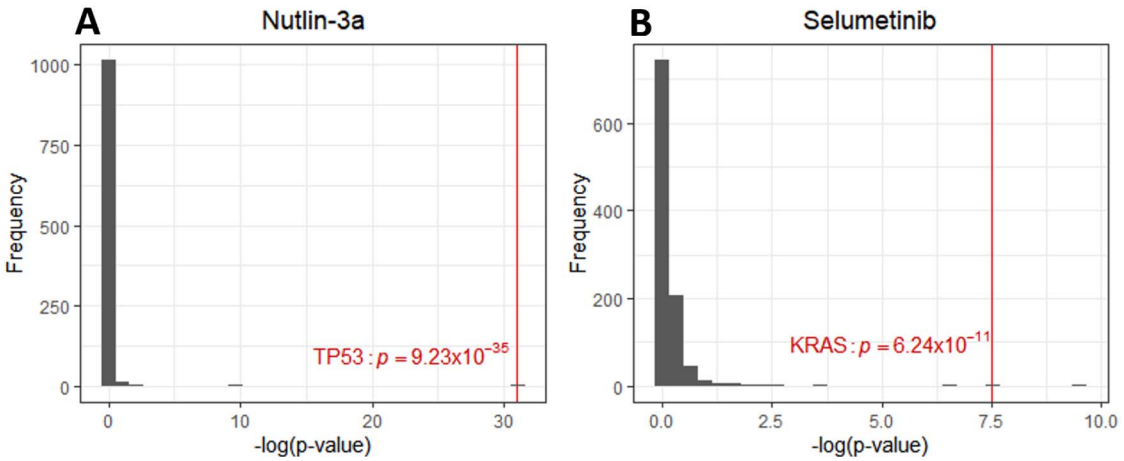

Figure 4. IDWAS in TCGA patient data. (**A**, **B**) Using the oncoPredict calcphenotype function, imputed sensitivities were generated for each patient against all the drugs in CTRP. The IDWAS function then allowed for testing associations between patient cancer gene mutations and patient imputed drug response values. For nutlin-3a (**A**) and selumetinib (**B**), histograms that show the significance (*P*-value) frequency across all the gene–drug associations are plotted.

Key Points

- We present here an R package to bridge *in vitro* drug screening with *in vivo* drug and biomarker discovery.
- oncoPredict enables generation of predictive models of drug response that can be used as a virtual screen for patient response to medications.
- Furthermore, oncoPredict enables biomarker discovery by imputing drug response in large cancer molecular datasets.
- oncoPredict enables drug-specific biomarker identification by correcting for GLDS shared among the population.

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Availability

Please check https://github.com/maese005/oncoPredict and https://osf.io/c6tfx/ for data.

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